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Graduation Project Dissertation

to obtain the National Diploma of Applied License in Development of Embedded and Mobile Systems

Dev and Deployment of an automated real-time monitoring system for TESCA group

Carried out by :

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Defended on 00/00/2026, in front of the committee members :

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End of studies internship completed at

Host company tesca

Client company TS2I Technique Solution et Innovation Industriel

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Graduation Project in Development of Embedded and Mobile Systems	
Made by: Fedi Amdouni Firas Ouesleti	**Batch:** 2025-2026
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Summary:

In modern manufacturing, accurate production monitoring is essential for optimizing efficiency. However, many facilities face unreliable traceability of machine stoppages and their root causes, hindering precise Overall Equipment Effectiveness (OEE) calculation and performance analysis. This project addresses this challenge by designing and deploying an automated, real-time industrial monitoring system. Built on a multi-layer architecture, the solution integrates PLCs via Modbus TCP for data acquisition, Node-RED for real-time processing, NestJS for backend management, and a custom HMI for operator interaction. The system automatically detects stoppages, enables real-time cause logging, and centralizes production data in a MySQL database, while analytical dashboards provide dynamic KPI visualization. Implementation results demonstrate complete machine event traceability, automatic OEE calculation per shift and day, and a significant reduction in manual entry errors and undetected production losses. The system successfully enhances production transparency and decision-making, with future work planned for scaling across additional production lines.

Keywords: Industrial monitoring, OEE, Modbus TCP, PLC, Node-RED, NestJS, HMI, machine stoppages, MySQL database

Résumé:

Dans l'industrie moderne, une surveillance précise de la production est essentielle pour optimiser l'efficacité. Cependant, de nombreuses installations font face à une traçabilité peu fiable des arrêts machine et de leurs causes profondes, entravant le calcul précis de l'Efficacité Globale des Équipements (OEE) et l'analyse des performances. Ce projet relève ce défi en concevant et déployant un système de surveillance industrielle automatisé et en temps réel. Construite sur une architecture multi-couches, la solution intègre des automates programmables (PLC) via Modbus TCP pour l'acquisition des données, Node-RED pour le traitement en temps réel, NestJS pour la gestion backend, et une interface homme-machine (IHM) personnalisée pour l'interaction des opérateurs. Le système détecte automatiquement les arrêts, permet l'enregistrement en temps réel des causes, et centralise les données de production dans une base de données MySQL, tandis que des tableaux de bord analytiques fournissent une visualisation dynamique des indicateurs de performance (KPI). Les résultats de l'implémentation démontrent une traçabilité complète des événements machine, un calcul automatique de l'OEE par équipe et par jour, et une réduction significative des erreurs de saisie manuelle et des pertes de production non détectées. Le système améliore avec succès la transparence de la production et la prise de décision, avec des travaux futurs prévus pour l'extension à d'autres lignes de production.

Mots-clés: Surveillance industrielle, OEE, Modbus TCP, PLC, Node-RED, NestJS, IHM, arrêts machine, base de données MySQL

Dedication

First and foremost, I express my deepest gratitude to Allah for His infinite blessings, wisdom, and strength throughout this journey. His guidance has been my compass in overcoming challenges and achieving this milestone.

To my favorite cheerleaders in life,

To my parents,

The architects of my dreams and the guardians of my laughter. Your love has painted my world with vibrant hues of happiness and warmth. This piece is just a little way to show how much I appreciate all the sacrifices you've made and the endless love you've given me.

To my brother,

The partner in crime,with your playful banter and unwavering support, you've made every moment an adventure worth living. This dedication is a reminder of the special bond we share and the joy you bring into my life.

To my dear friends,

The stars that light up my sky on the darkest of nights. Your friendship is my greatest treasure. This work is dedicated to you as a symbol of our enduring bond and the countless memories we've yet to create.

May this dedication serve as a humble token of my gratitude to all those who have touched my life with their kindness, wisdom and love.

Fedi Amdouni

Dedication

First and foremost, I would like to thank God for all His blessings upon me and my family. For the strength, patience, and courage He granted me each day to pursue my goals and accomplish this final year project.

I dedicate this humble work to:

To my beloved parents,

Your unwavering support, endless sacrifices, and constant prayers have been the foundation of my success. May Allah grant you long life, good health, and the happiness you truly deserve.

To my dear sister,

For your encouragement, understanding, and the joy you bring to my life. Your belief in me has been a constant source of motivation.

To my partner, Fedi Amdouni,

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To all my friends and mentors,

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Firas Ouesleti

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Fedi Amdouni & Firas Ouesleti

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General Introduction

Chapter 1

Preliminary analysis

Introduction

The first chapter of this report aims to provide an overview of the host and the client companies, present a case study highlighting the problem to be addressed, and introduce the modeling language used for the project. This chapter sets the stage for the subsequent sections by establishing the context, significance, and scope of the actions undertaken in the next chapters.

1.1 Presentation of the enterprises

1.1.1 TESCA

Tesca Group is a family-run company with a background in textiles that has successfully ventured into the automotive industry, specifically automotive seating. Emulating the approach of a French automotive manufacturing house, Tesca Group meticulously crafts each piece, from its atelier to its global production sites [26]. The company strives to embody the same creative audacity and prides itself on being a partner to major automotive manufacturers like:

Figure 1.1: Tesca's partners

With a clientele consisting of demanding visionaries who shape the future of mobility, Tesca Group positions itself as a reliable and flexible partner, accompanying its clients in their pursuit of innovation and competitiveness. The company believes that adopting a long-term and eco-friendly vision is crucial for sustainable growth, constantly pushing for excellence rather than settling for mediocrity [26].

Sustainability is a concrete and integral part of Tesca Group's approach. The company is committed to sustainable development and actively incorporates environmental and social values into its products, services, and activities within the automotive industry. Since 2004, Tesca Group has been dedicated to controlling its ecological footprint, and the executive management maintains and expands upon these efforts. The company adheres to the values and principles of the Global Compact through its Ethics Charter, to obtain ISO 14001 certification for all its sites [26].

Tesca Group's environmental policy revolves around five major directions: behaving in an environmentally responsible manner, optimizing and controlling the industrial waste, optimizing energy consumption and natural resources, participating in the reduction of the ecological footprint throughout the design and production processes, and ensuring compliance with environmental regulations and requirements [26].

Innovation is a driving force within Tesca Group, with a constant emphasis on generating new ideas and executing them effectively. The company's research programs focus on key themes in the automotive industry, such as weight reduction, optimized part design, and ecological footprint expertise, while prioritizing user comfort, functionality, and safety. Tesca Group holds numerous patents in the conception, design, and manufacturing of automotive parts and seat components, including headrests, armrests, upholstery, padding, and "smart textiles" [26].

Industrial excellence is a cornerstone of Tesca Group's strategy, whether through its local production sites or research centers. The company strives to provide reliable processes that meet the evolving market expectations worldwide. Tesca Group empowers its teams through a culture of continuous improvement, with competitiveness, efficiency, and exceptional service forming the foundation of its industrial success. SPRINT (Tesca's Industrial Production System) is employed to engage collaborators at all levels, aiming to standardize procedures and ensure perpetual progress in industrial performance. Shared values among Tesca Group employees include excellence, self-discipline, commitment, autonomy, and pragmatism [26].

Tesca Group's Quality policy aligns with five major directions: meeting commitments to conformity, cost, and deadlines; aligning industrial and purchasing performance with the highest quality criteria; standardizing existing processes to offer innovative products; prioritizing process control and prevention; and continuously enhancing the skills of its teams [26].

In terms of its history, Tesca Group has a notable timeline of achievements. Starting in 1960 with the provision of Citroën 2CV roofs, the company went on to pioneer and apply the Foam In Place technology in 1978. In 1983, Tesca Group established its Design Studio in Paris, followed by the creation of the CERA research and development center in Reims in 1993. By the year 2000, the company had expanded its production capabilities worldwide. In 2016, Tesca Group underwent a significant transformation, becoming Trèves TSC [26].

Overall, Tesca Group has established itself as a reputable partner to major automotive manufacturers, driven by a commitment to sustainability, innovation, industrial excellence, and quality. Through its diverse range of products and services, the company aims to shape the future [26].

1.1.2 TS2I

TS2I is a leading company in the field of specialised machines fabrication, automation, industrial maintenance, and Tampographique service. With a strong commitment to quality, innovation, and customer satisfaction, TS2I has established itself as a trusted partner for various industries. The following provides an overview of TS2I and highlights its expertise in fabrication machines, automation, industrial maintenance, and pad printing services.

- **Fabrication of specialised machines:** TS2I specializes in the design, development, and production of fabrication of specialised machines. These machines are tailored to meet the unique requirements of clients in diverse industries such as automotive, aerospace, electronics, and more. With a team of experienced engineers and technicians, TS2I leverages cutting-edge technologies to deliver custom fabrication machines that optimize productivity, efficiency, and quality. Whether it's automated assembly lines, robotic systems, or specialized machining equipment, TS2I ensures that each machine is built to the highest standards.

- **Automation:** Automation plays a crucial role in enhancing productivity and reducing operational costs. TS2I offers comprehensive automation solutions to help businesses automate their processes. From simple control systems to complex integrated automation solutions, TS2I designs and implements tailored automation solutions based on clients' requirements. Their expertise in PLC programming, HMI design, robotics, and motion control allows them to deliver efficient and reliable automation solutions that streamline operations and improve overall efficiency.

- **Industrial maintenance:** TS2I understands the importance of maintaining industrial equipment to ensure optimal performance and minimize downtime. The company offers comprehensive maintenance services to address the unique needs of its clients. Whether it's preventive maintenance, troubleshooting, or equipment repair, TS2I's skilled technicians and engineers are well-equipped to handle a wide range of industrial machinery. By providing regular maintenance, TS2I helps clients avoid costly breakdowns, increase equipment lifespan, and maximize productivity.

- **Tampographique service:** In addition to its fabrication machines and automation expertise, TS2I also offers a specialized service in tampography, or pad printing. Tampography is a versatile printing technique used for applying designs or logos onto various surfaces, including plastics, metals, ceramics, and glass. TS2I's pad printing service ensures high-quality and precise printing, providing clients with branding solutions that are durable and visually appealing. Whether it's product customization, promotional items, or industrial labeling, TS2I's pad printing service offers reliable and efficient solutions.